

# Ethan Ernesto Lopez

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## TECHNICAL SKILLS

Statistical Tools & Languages – SQL | Python | R | C++ | Java | Git | GitHub | Docker | Linux | Pivot Tables | Matplotlib | Dplyr  
Libraries & Platforms – Pandas | NumPy | Scikit-Learn | Jupyter | Google Colab | Alteryx Cloud | Tableau | StatsModels | Excel  
Data – Hypothesis Testing | EDA | Regression | Clustering | Machine Learning | Social Network Analysis | Classification Models

## PROFESSIONAL EXPERIENCE

**Research Assistant – Social Network Analysis (MR-QAP)** | *Chapman University* March 2026 – Present

- Evaluate collaboration network models for 240+ first-year STEM students in 11 classrooms via Python and MRQAP.
- Enhance datasets in cleaning and transformation methods, improving analytical precision and validation by 20%.
- Apply 4+ statistical methods including MR-QAP, ANOVA, and density to assess team stability and quality interactions.
- Develop more than 7 network visualizations and graphs explaining 64.4% variation in team formation outcomes.

**Volunteer -> Senior Leadership Instructor** | *UTO Taekwondo – Cypress, CA* March 2018 – Present

- Began as a volunteer and transitioned into a paid role, providing office support and handling customer inquiries.
- Supervise 15+ youth per shift in self-defense, manage entry-level team members, and oversee studio operations.

**Academic Tutor (Student Tutor)** | *Chapman University* September 2024 – May 2026

- Tutored 5+ students quarterly for 10 STEM and business topics in statistics, data analysis, and quantitative modeling.
- Expanded on techniques in regression, hypothesis testing, and data interpretation to improve problem-solving skills.
- Utilized over 13 statistical procedures in Python (NumPy, Pandas), R, and Excel, analyzing student feedback metrics.

**Data Entry/Admin Assistant – Office of Accreditation & Assessment** | *Chapman University* June 2025 – August 2025

- Maintained accuracy in graduate databases with 50+ students, validating survey metrics in accreditation assessments.
- Conducted research and enrollment trend analysis on more than 70 alumni to support counseling program outcomes.
- Leveraged visualizations using Excel and Qualtrics to improve stakeholder understanding of internships by 25%.

**Teaching Assistant – Summer Engineering Academy (Arduino Cars)** | *Chapman University* June 2025 – July 2025

- Provided technical assistance to 40+ students in Arduino programming, circuit design, and robot hardware assembly.
- Guided workshops in autonomous robot vehicle debugging methods to ensure 80% deployment success per section.
- Managed more than 20 materials and maintained documentation using Office tools to support program operations.

## RELEVANT PROJECTS & EXPERIENCE

**Spotify Tune Success Analytics Dashboard** – *Business Intelligence & Data Visualization* April 2026 – May 2026

- Conducted analysis on over 130,000 Spotify songs with 17 audio metrics to identify key factors in streaming success.
- Built 10+ interactive Tableau visuals, enabling users to gain insights for 5+ correlations in audio song performances.
- Applied methods in hypothesis tests to reveal over 6 significant song popularity variables (energy, danceability, etc.).

**Predicting Physical Activity From Wearable Health Metrics** – *Machine Learning* April 2026 – May 2026

- Performed data analysis on over 6,000 health observations, observing physiological predictors for physical activities.
- Built machine learning workflows for data preprocessing, tuning 6+ hyperparameters for classification algorithms.
- Achieved an ROC-AUC of 0.92 with 76% model accuracy, feature importance analysis identifying 7 key classifiers.

**Large-Scale Customer Segmentation Analysis** – *Big Business Predictive Modeling* November 2025 – December 2025

- Collaborated with 2 others to integrate Jupyter/Colab machine learning with over 9 million online booking records.
- Implemented feature engineering and data transformation to apply clustering analysis for 6 customer segmentations.
- Trained 5+ regression and classification models, designing Matplotlib visuals to present to a non-technical audience.

## EDUCATION

Chapman University, Orange, CA  
**Bachelor of Science in Data Science, Summa Cum Laude**  
**Minor in Business Administration**

May 2026  
*Program Honors Graduate*  
3.92 GPA

## ACADEMIC AWARDS AND LEADERSHIP

**Fowler School of Engineering Outstanding Senior Award – Data Science** 2026  
**Outstanding Tutor Certificate of Appreciation** 2026  
**Presidential Scholarship Recipient** 2022–2026  
**Provost's List** 2022–2025